Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \mathrm{P}_{j,j+i} = &\sum_{m=i+1}^{M-j} \binom{M-j}{m} \mathbb{P}_{\mathrm{TX}}^{m} \left(1-\mathbb{P}_{\mathrm{TX}}\right)^{M-j-m} \\ &\times \frac{M-j-i}{M-j} m \cdots (m-i+1) \mathbb{P}_{K}^{m} \gamma_i \gamma_{m,i} \\ &+ \binom{M-j}{i} \mathbb{P}_{\mathrm{TX}}^{i} \left(1-\mathbb{P}_{\mathrm{TX}}\right)^{M-j-i} \frac{M-j-i}{M-j} i! \mathbb{P}_{K}^{i} \gamma_i, \end{aligned}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} Q(x) &= \sum_{x/\log x \leq p \leq x-\sqrt{x}} \pi(x-p) + O\Bigg(\sum_{p \leq x/\log x} \pi(x) + \sum_{x-\sqrt{x} \leq p \leq x} x \Bigg) \\ &= \sum_{x/\log x \leq p \leq x-\sqrt{x}} \pi(x-p) + O\Bigg(\pi(x) \pi\left(\frac{x}{\log x}\right) + x\sqrt{x} \Bigg) \\ &= \sum_{x/\log x \leq p \leq x-\sqrt{x}} \pi(x-p) + O\left(\frac{x^2}{\log^3 x}\right). \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \partial_{\hat{\xi}} u(\hat{\xi},0) - \partial_{\hat{\xi}} w(\hat{\xi},0) &\leq \partial_{\hat{\xi}} u(\hat{\xi},0) - \partial_{\hat{\xi}} w(0,0) + [\partial_{\hat{\xi}} \hat{w}]_{0,\beta} |\hat{\xi}|^{\beta} \\ &= \left[U_{\beta}(1+\beta) \cos\left((1+\beta)\frac{\pi}{2}\right) + [\partial_{\hat{\xi}} \hat{w}]_{0,\beta} \right] |\hat{\xi}|^{\beta}, \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$
\begin{aligned}
&II+III\to 0;\\
&\quad IV=\int_{d(x,p)\geq\rho}(\frac{\rho^2}{4\epsilon}-n)e^{-\frac{\rho^2}{4\epsilon}-C}\to 0;\\
&\quad\quad I=e^{-C}\int_{|y|\leq\frac{\rho}{\sqrt{\epsilon}}}(\frac{|y|^2}{2}-n)(2\pi)^{-n/2}e^{-|y|^2/4}(1+O(\epsilon|y|^2)dy\\
&\quad\quad\quad\to\int_{\mathbb{R}^n}(\frac{|y|^2}{2}-n)(2\pi)^{-n/2}e^{-|y|^2/4}dy=0.
\end{aligned}
$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$
\begin{aligned}
\frac{1}{\omega(B)}\|(b-b_B)f_1\|_{L^s_\omega}^s &= \frac{1}{\omega(B)}\int_{5B}|b(y)-b_B|^s|f|^s d\omega(y)\\
&\leq\Big(\frac{1}{\omega(B)}\int_{5B}|b(y)-b_B|^{ss_1'}d\omega(y)\Big)^{\frac{1}{s_1'}}\Big(\frac{1}{\omega(B)}\int_{5B}|f|^{ss_1}d\omega(y)\Big)^{\frac{1}{s_1}}\\
&\lesssim \ell(B)^{\beta s}\|b\|_{\Lambda^\beta}^s M_{ss_1}^s(f)(x)
\end{aligned}
$$